

Data 102, Fall 2026

Homework 1

Drug Testing

1. Suppose a particular drug test is 99% sensitive and 98% specific ([Here](#) is a Wikipedia link for a refresher on the terminology). The null hypothesis H_0 is that the subject is not using the drug. Assume a prevalence of $\pi_1 = 0.5\%$, i.e. only 0.5% of people use the drug. Consider a randomly selected individual undergoing testing. Rounding to the nearest three significant figures, find:
 - (a) the probability of testing positive given H_0 .
 - (b) the probability that they are not using the drug given they test positive.
 - (c) the probability of testing positive a second time given they test positive once. You may assume the two tests are statistically independent given drug user status.

Distinguishing Exponentials

2. Suppose we have a waiting time $T \sim \text{Exponential}(\lambda)$ and wish to test

$$H_0 : \lambda = c \quad \text{vs} \quad H_1 : \lambda = 2c$$

for some $c > 0$. In this question, you'll use the *likelihood ratio test* (LRT) to compare these two hypotheses. The LRT considers the ratio of the two density functions f_1 and f_0 under the alternative and null respectively:

$$\text{LR}(T) = \frac{f_1(T)}{f_0(T)},$$

and rejects H_0 when $\text{LR}(T)$ is greater than some threshold η .

We use this test because of the *Neyman-Pearson lemma*, which states that the likelihood ratio test is the most powerful test (in other words, it has the highest power, or TPR) of significance level α . That is, out of all possible tests of H_0 vs H_1 with $\text{FPR} = \alpha$, the likelihood ratio test has the highest TPR.

Hint: This question will be easier to solve after Discussion 2. You may also find it helpful to brush up on computing probabilities involving continuous random variables. [Prob 140 textbook](#), [Chapter 15](#) provides a helpful refresher.

- (a) Compute $\text{LR}(T)$ explicitly in terms of c .
- (b) Let α be our false positive rate ($0 < \alpha < 1$). Compute the value of the threshold η so that the FPR of the test is equal to α . We say that such a test has *significance level* α . Your answer should be expressed in terms of any or all of α and c .
Hint: start by expressing the FPR as a conditional probability, then connect it to the LRT decision rule and the densities f_0 and f_1 .
- (c) What is the TPR of this test? This is also known as the test's *power*. Your answer should be expressed in terms of α and c .
- (d) Generalize parts (a) - (c) to $H_1 : \lambda = a \times c$ for arbitrary $a \geq 1$. Make a plot showing the TPR of the test as a function of desired FPR, α , for a ranging from 1 to 10.
- (e) Use your plot, or formula, from part (d) to answer each question below. Then, explain why each answer is a natural consequence of the problem setting (i.e. explain how you could have anticipated each answer before performing the problem):
 - i. Is the TPR a monotonic function of the FPR? If so, is it increasing or decreasing in the FPR?
 - ii. Is the TPR a concave or convex function of the FPR?
 - iii. What is the TPR when $\alpha = 0$ and when $\alpha = 1$?
 - iv. Holding α fixed, how does the TPR depend on a (increasing, decreasing, constant)? How does the TPR behave in the limiting cases $a = 1$ and $a = \infty$.
 - v. Does the TPR for the LRT depend on the value of c ? *Hint: can you relate the problem with arbitrary c to the problem with $c = 1$ by changing units for T ?*

Fairness in Binary Prediction

3. *Fairness* is a hotly debated topic and active area of study in machine decision-making. In this problem, you will show that two reasonable notions of fairness are mutually incompatible with accuracy, and will consider the consequences of that conclusion. The example is inspired by the paper “Inherent trade-offs in the fair determination of risk scores” [1]. **Please read the full set-up before attempting the problems. You will not be able to answer the problems without reading first.**

Suppose that a data scientist is asked to design a prediction tool that takes in a job candidate's resume, and predicts whether or not the candidate will succeed in an interview. The data scientist aims to produce a tool that is both accurate and fair.

We'll use the following notation:

- 1. $i \in \{1, 2, \dots, N\}$ indexes each candidate. I will denote a uniform random individual: $I \sim \text{Uniform}(1, 2, \dots, N)$.
- 2. $y_i \in \{0, 1\}$ is an indicator for whether candidate i , if interviewed, will succeed, i.e. pass the interview.

3. **Protected Classes:** $\mathcal{C} = \{C_1, C_2, \dots, C_n\}$ is a collection of protected classes (e.g. age, gender, race, etc.), where each $C_j \subset \{1, 2, \dots, N\}$ contains a subset of the candidates all belonging to the same class. For example, the first protected class C_1 might consist of all individuals who are female, Black, and 40-50 years old; the second protected class C_2 might contain all candidates who are male, American Indian/Alaska Native, and 50-60 years old; and so on. You may assume that the classes are disjoint.
4. p_j is the probability that a random individual from class C_j would pass the interview:
 $p_j = \mathbb{P}(y_I = 1 \mid I \in C_j)$
5. **Predictive types:** $\mathcal{T} = \{T_1, T_2, \dots, T_m\}$ is a collection of disjoint candidate types used for prediction. Each $T_k \subset \{1, 2, \dots, N\}$ contains a subset of candidates who the prediction tool will treat as interchangeable. It will assign all individuals of a given type the same probability of passing their interview. For example, suppose the tool tracks education level achieved. Then the types could be:
 - T_1 contains all individuals who did not graduate high school;
 - T_2 contains all high school graduates (who did not earn any further degrees);
 - T_3 contains all college graduates (who did not earn any further degrees); and
 - T_4 contains all applicants with a graduate or professional degree.
6. $t_k = |T_k|$ is the number of individuals of type k . In the example above, t_4 is the number of applicants with graduate or professional degrees
7. $k(i)$ is the index corresponding to the type of individual i . In the example above, $k(77) = 3$ means that applicant 77 has a college degree and no further education.
8. The prediction tool assigns each type T_k a “score”, s_k , then recommends individuals of type k with probability equal to their score s_k .
9. $z_i \in \{0, 1\}$ is the decision returned by the tool for individual i . If $z_i = 1$, then the tool recommends the individual. Specifically, $z_i \sim \text{Bernoulli}(s_{k(i)})$.

The data scientist aims to choose the scores, $\{s_1, \dots, s_m\}$, so that the score assigned to each individual closely matches their chance of succeeding in the interview **without treating protected classes differently**. So, they adopt three objectives:

1. **Calibration:** The conditional probability that the tool *recommends* an individual should equal the conditional probability that the individual *would succeed* in their interview, given their type and class membership, for all types and classes:

$$\mathbb{P}(z_I = 1 \mid I \in C_j, I \in T_k) = \mathbb{P}(y_I = 1 \mid I \in C_j, I \in T_k) \text{ for all } j, k. \quad (1)$$

This is an **accuracy** objective, since it matches the tool’s recommendation probabilities (scores) to the true success probabilities for each predictive type and each protected class.

2. **Matching True Positive Rates:** The true positive rate for a uniformly drawn individual *should not* depend on their class:

$$P(z_I = 1 \mid y_I = 1, I \in C_j) = P(z_I = 1 \mid y_I = 1, I \in C_{j'}) = \text{TPR for all } j, j'. \quad (2)$$

This is a **fairness** objective: when we compare individuals who would pass the interview, it says that the tool should be equally accurate on all of them, even if they're from different protected classes.

3. **Matching True Negative Rates:** The true negative rate for a uniformly drawn individual does not depend on their class:

$$P(z_I = 0 \mid y_I = 0, I \in C_j) = P(z_I = 0 \mid y_I = 0, I \in C_{j'}) = \text{TNR} \text{ for all } j, j'. \quad (3)$$

This is also a **fairness** objective: when we compare individuals who would *not* pass the interview, it says that the tool should be equally accurate on all of them, even if they're from different protected classes.

In this question, you'll show that, except in very special cases, no prediction tool can satisfy all three objectives at once.

- (a) By marginalizing over predictive types, show that, if the prediction tool is well calibrated (i.e., it meets the calibration objective), then:

$$\mathbb{P}(z_I = 1 \mid I \in C_j) = p_j. \quad (4)$$

Hint: you should not make any assumptions about either fairness objective for here.

- (b) Show that if the tool satisfies both of the fairness objectives (matching TPR and matching TNR), then for any protected class C_j ,

$$(2 - (\text{TPR} + \text{TNR}))p_j = \text{FPR} \text{ for all } j. \quad (5)$$

Hint: Expand the left hand side of equation (4) by using the law of total probability, partitioning over the possible outcomes when candidate I is interviewed. Then, use the definitions of TPR, FPR, FNR and apply the fairness constraint.

- (c) Show that equation (5) can only be solved in one of two cases. Either, the probability that an individual succeeds in their interview does not depend on their class ($p_j = p_{j'}$ for all j, j'), or, the tool is perfect and never makes any mistakes (i.e. perfect prediction). If the tool is well-calibrated, is perfect prediction possible for arbitrary p_j ? If not, what are the only situations when perfect prediction is possible?

Hint: think about what condition(s) must be true for each case where perfect prediction is possible. What does that imply about each individual?

- (d) What do *you* think this result implies about fairness in binary decision-making? Think about the difference between enforcing constraints exactly and approximately, whether the chosen definitions of fairness and accuracy could be reasonably adapted or replaced, and, what other judgments the data scientist must make if they cannot satisfy all of their goals at once.

Hint: "Fairness is impossible" is not a correct answer, and won't receive any credit!

- (e) In the example above, we assumed that the predictive types corresponded to maximal degree achieved. Did that specific choice matter, or are calibration, matching true

positive, and matching true negative rates incompatible for any definition of types? Is it possible to align the set of types $\mathcal{T}$ to the protected classes $\mathcal{C}$ so that the three constraints are compatible?

References

- [1] Kleinberg, Jon and Mullainathan, Sendhil and Raghavan, Manish. *Inherent trade-offs in the fair determination of risk scores*. arXiv preprint arXiv:1609.05807, 2016.
- [2] Bradley Efron. *Large-scale inference: empirical Bayes methods for estimation, testing, and prediction*. Cambridge University Press, 2012.